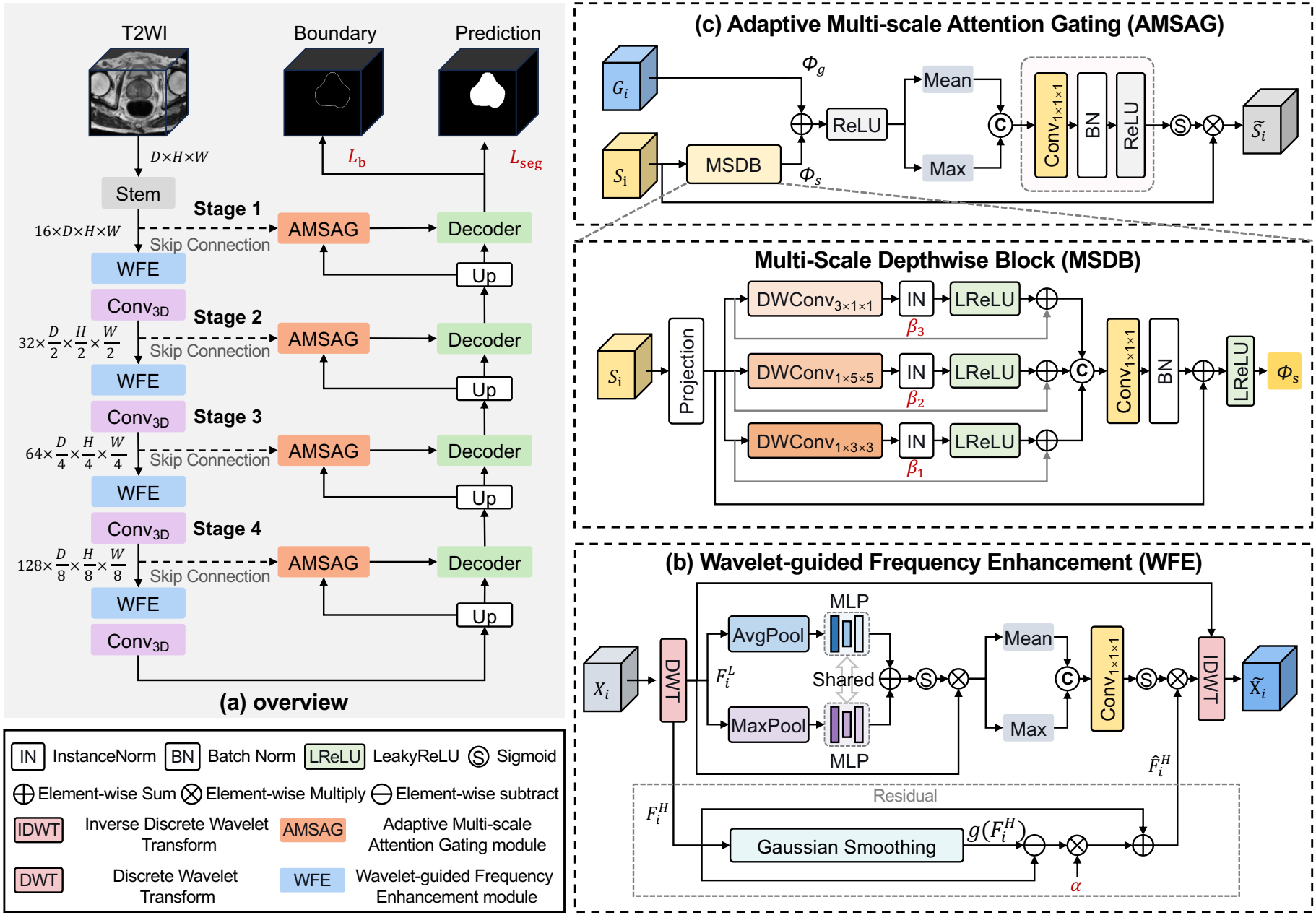

T2WI
Boundary
Prediction
D×H×W
L_b
L_seg
Stem
16×D×H×W
Stage 1
Skip Connection
AMSAG
Decoder
WFE
Up
Conv3D
32×D/2×H/2×W/2
Stage 2
64×D/4×H/4×W/4
Stage 3
128×D/8×H/8×W/8
Stage 4
(a) overview
(c) Adaptive Multi-scale Attention Gating (AMSAG)
G_i
S_i
MSDB
Φ_g
Φ_s
ReLU
Mean
Max
C
Conv1×1×1
BN
ReLU
S
S̃_i
Multi-Scale Depthwise Block (MSDB)
Projection
DWConv3×1×1
DWConv1×5×5
DWConv1×3×3
IN
LReLU
β_3
β_2
β_1
Conv1×1×1
BN
LReLU
Φ_s
(b) Wavelet-guided Frequency Enhancement (WFE)
X_i
DWT
F_i^L
AvgPool
MaxPool
MLP
Shared
MLP
Mean
Max
C
Conv1×1×1
S
IDWT
X̃_i
F̂_i^H
Residual
F_i^H
Gaussian Smoothing
g(F_i^H)
α
IN InstanceNorm
BN Batch Norm
LReLU LeakyReLU
S Sigmoid
Element-wise Sum
Element-wise Multiply
Element-wise subtract
IDWT Inverse Discrete Wavelet Transform
AMSAG Adaptive Multi-scale Attention Gating module
DWT Discrete Wavelet Transform
WFE Wavelet-guided Frequency Enhancement module